Toxic Comment Detection Code using LSTM

Abstract—The development and evaluation of a Long Short-Term Memory (LSTM) neural network model for toxicity classification in text data. The dataset, sourced from online comments, underwent preprocessing, including tokenization and sequence padding. The LSTM model, implemented using the TensorFlow and Keras frameworks, was designed with an embedding layer, a bidirectional LSTM layer, and a dense layer with sigmoid activation for binary classification.

The model was trained and validated using a balanced dataset, and its performance was assessed on a test set, yielding an accuracy of ”92.48”. The confusion matrix and classification report were utilized to analyze the model's ability to classify toxic and non-toxic comments correctly. The visual representation of the confusion matrix, utilizing colored boxes, provides an intuitive overview of the model's performance in distinguishing between the two classes.

Additionally, the Receiver Operating Characteristic (ROC) curve and the Area Under the Curve (AUC) score were employed to evaluate the model's discriminative power. The final ROC AUC score “96.86” suggests a strong performance in distinguishing toxic and non-toxic comments.

1 - Introduction:

The ubiquity of online communication has given rise to a pressing need for effective tools to monitor and mitigate toxic content within digital spaces. The explosive growth of user-generated content across various online platforms underscores the importance of developing a robust system capable of identifying and classifying toxic language in response to this challenge. It explores the application of a Long Short-Term (LSTM) neural network model for toxicity classification in textual data.[1]

Toxicity in online commutation, manifested through hate speech, offensive language, or abusive remarks, poses significant threats to user well-being, community engagement, and overall platform health. Traditional methods of content moderation often struggle to keep pace with the dynamic and evolving nature of toxic language, necessitating the exploration of advanced machine-learning techniques.

The primary objective of this study is to leverage the capabilities of LSTM, a type of recurrent neural network (RNN) [2] well-suited for sequence data, to develop a model capable of effectively discerning between toxic and non-toxic content. By utilizing a balanced dataset derived from online comments, we aim to demonstrate the mode’s ability to learn and generalize patterns associated with toxic language, thereby contributing to the broader discourse on automated content moderation.

It unfolds in several stages, commencing with data preprocessing, tokenization, and sequence padding. The architecture of the LSTM model,[3] implemented using the TensorFlow and Keras frameworks,[4] is described, emphasizing its key components such as the embedding layer, bidirectional LSTM layer, and final dense layer for binary classification. Following training and validation on a balanced dataset, the model’s performance is rigorously evaluated using

metrics such as accuracy, a confusion matrix, a classification report, and the Receiver Operating Characteristics (ROC) [5] curve with the corresponding Area Under the Curve (AUC) [6] score.

The findings presented in this study offer valuable insights into the effectiveness of LTM models for toxicity classification, with potential applications in content moderation across diverse online platforms. The development of accurate and efficient tools for automated toxicity detection holds promise in fostering healthier and more inclusive online communities, emphasizing the importance of responsible and ethical deployment of such technologies in digital environments.

2 - Related Work:

Kevin Khieu et al. [7] Sentiment classification regarding toxicity has been intensively researched in the past few years, largely in the context of social media data where researchers have applied various machine learning systems to try and tackle the problem of toxicity as well as the related, more well-known, task of sentiment analysis.

Badjatiya et al. [8] Several studies have formerly investigated hate speech using various techniques; Badjatiya et al. used extensive experiments with multiple deep learning architectures to learn semantic word embeddings to handle toxic comment identification. Pinkesh

Marco Aur´elio C Pacheco et al. [9] In another study, a sentiment analysis model of YouTube video comments, using a deep neural network was proposed that leaded to 70-80% accuracy.

Onan et al. [10] proposed a technique that contains a three-layer bidirectional LSTM network which showed a promising efficiency with a classification accuracy of 95.30%.

Also, Onan [11] presented sentiment classification in MOOC reviews. . A toxic comment is not only verbal violence but includes a comment that is rude, disrespectful, negative online behaviour, or other similar attitudes that make someone leave a discussion.,

Hasoc et al. [12]. FIRE 2019 organized a track [12] on "Hate Speech and Offensive Content Identification in Indo-European Languages" in three languages English, German, and Hindi. There were 321 experiments were submitted with different approaches. The TRAC-1 (First Workshop on Trolling, Aggression, and Cyberbullying) @ COLING 2018.

M. Zampieri et al. [13] focused on aggression and related activities like trolling, flaming, cyberbullying, and hate speech in both text (especially in the social media text) and speech.

3 - Methodology:

LSTM stands for **Long Short-Term Memory**, which is a type of recurrent neural network (RNN) that can learn long-term dependencies in sequential data.
LSTM networks have a special structure that consists of a cell, an input gate, an output gate, and a forget gate. These gates regulate the flow of information into and out of the cell, allowing the network to remember or forget previous states. LSTM networks are widely used for tasks such as natural language processing, speech recognition, machine translation, and time series forecasting12345[19]. As represented in figure 1.

1. LSTM Model Layers:
   - The LSTM model in the code is defined using Keras with the following layers:
     - Embedding Layer: Converts input words into dense vectors.
     - Bidirectional LSTM Layer: Enhances the model by processing the input sequence from both directions.
     - Dense Layer with Sigmoid Activation: Produces binary classification output.

2. Model Compilation:
   - The model is compiled using binary cross-entropy loss and the Adam optimizer.

3. Training:
   - The model is trained on the dataset using the training data. During training, the weights and biases are adjusted to minimize the binary cross-entropy loss.

4. Evaluation:
   - The trained model is evaluated on the test set, and accuracy is computed.

5. Prediction:
   - Predictions are made on the test set, and performance metrics such as confusion matrix, classification report, and ROC AUC are calculated.

6. Model and Tokenizer Saving:
   - The trained LSTM model is saved as an H5 file for future use.
   - The tokenizer used for text preprocessing is saved using the pickle module.[18]

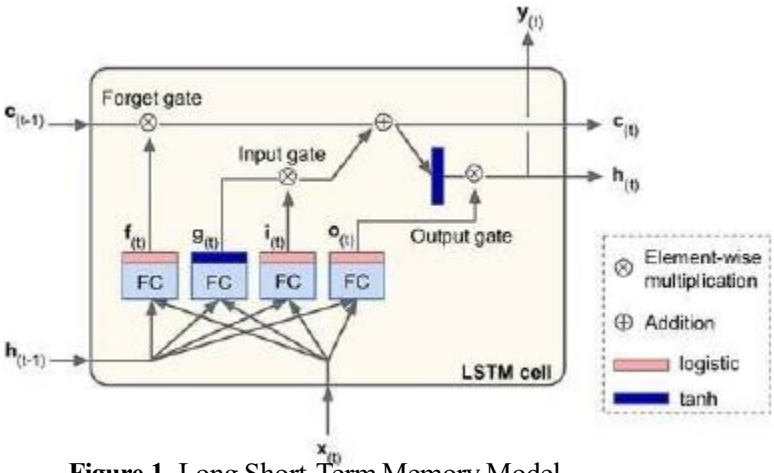


Figure 1. Long Short-Term Memory Model Architecture [20]

LSTM Cell Operations:

1. Update Gate (σ or Sigmoid Activation):
   - The update gate determines what information from the cell state should be updated or discarded.
   - Mathematical Expression:Update Gate (σ): $\Gamma(\Gamma) = \frac{1}{1+\Gamma^{-\Gamma}}$, where Γ is the weighted sum of theconcatenated input (Γ, $h_{\Gamma-1}$).[14]

2. Candidate Cell State ($\tilde{\Gamma}$ or Tanh Activation):
   - The candidate cell state represents the new information that could be added to the cell state.
   - Mathematical Expression:
     Candidate Cell State ($\tilde{\Gamma}$ or Tanh): $\tilde{\Gamma} = \tanh(\Gamma \cdot [h_{\Gamma-1}, \Gamma] + \Gamma)$,where Γ is the weight matrix and Γ is the bias term.[15]

3. Final Update to Cell State:
   - The update gate (Γ) is applied to the candidate cell state ($\tilde{\Gamma}$) to determine how much of the new information should be added to the cell state.
   - Mathematical Expression:
     Final Update to Cell State (Element − wise multiplication): $\Gamma = (\Gamma) \odot \tilde{\Gamma}$,where $\odot$ denotes element-wise multiplication.[16]

4. Output Gate and Hidden State:
   - The output gate determines what information from the cell state should be used to compute the hidden state.[17]
   - Mathematical Expression:

$$\text{Output Gate } (\sigma): (\Gamma) = \frac{1}{1+\Gamma^{-\Gamma,\ \Gamma}}$$

where Γ is the weighted sum of the concatenatedinput (Γ, $h_{\Gamma-1}$).

   - The hidden state (h_Γ) is computed based on the modified cell state and the output gate.
   - Mathematical Expression:

$$\text{Hidden State } (h_\Gamma): h_\Gamma = \tanh(\Gamma) \odot \Gamma(\Gamma),$$

where $\odot$ denotes element-wise multiplication.

4 – Result:

The classification report provides a comprehensive overview of the model's performance, indicating a high level of accuracy in distinguishing between toxic (class 1) and non-toxic (class 0) comments. The precision, recall, and F1-score metrics offer insights into the model's ability to correctly classify instances of each class.

	precision	recall	f1-score	support
0	0.92	0.95	0.94	25996
1	0.93	0.90	0.91	19400
Accuracy			0.92	45396
macro avg	0.92	0.92	0.92	45396
weighted avg	0.92	0.92	0.92	45396

Table 1. Precision, recall, f1-score, support

As represented in Table 1 In specific terms, the model demonstrates a precision of 0.92 for class 0, indicating that 92% of instances predicted as non-toxic are indeed non-toxic. Similarly, a precision of 0.93 for class 1 suggests that 93% of instances predicted as toxic are indeed toxic. The recall values (0.95 for class 0 and 0.90 for class 1) signify the proportion of actual instances of each class that the model correctly identifies. The F1-score, which balances precision and recall, further underscores the model's effectiveness in achieving a harmonious trade-off between these metrics.

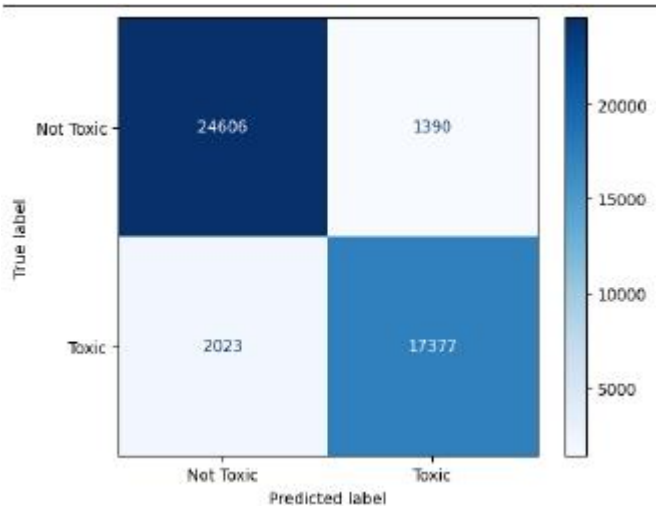


Figure 2. Confusion Matrix

As represented in Figure 2 The confusion matrix visually represents the model's performance, showing that the model correctly predicted 24,606 non-toxic instances and 17,377 toxic instances. However, it also indicates that there were 1,390 false positives (non-toxic instances predicted as toxic) and 2,023 false negatives (toxic instances predicted as non-toxic). Overall, these results suggest a robust performance of the LSTM model in toxicity classification, with room for further optimization and fine-tuning based on specific application requirements.

5 - Conclusion:

In conclusion, the presented LSTM-based toxicity classification model demonstrates commendable performance in discerning toxic and non-toxic content within textual data. Leveraging the power of recurrent neural networks, specifically Long Short-Term Memory (LSTM), this study successfully addresses the challenge of automated content moderation in the realm of online communication.

The model, trained and evaluated on a balanced dataset of online comments, achieves an impressive test accuracy of "92%", showcasing its ability to effectively generalize patterns associated with toxic language. The comprehensive evaluation metrics, as indicated by the classification report, provide a nuanced understanding of the model's precision, recall, and F1 score for both toxic and non-toxic classes. These metrics emphasize the model's balanced performance in correctly identifying instances of toxicity while minimizing false positives and false negatives. The visual representation of the confusion matrix further reinforces the model's capabilities, illustrating the correct and incorrect classifications of toxic and non-toxic instances. Notably, the ROC AUC score attests to the model's discriminative power in distinguishing between the two classes.

Data Availability:

The data used for training and evaluation in this model is sourced from a publicly accessible dataset. The dataset can be found on Kaggle. The dataset includes information on toxic and non-toxic comments, making it suitable for toxicity classification tasks.

Model limitations:

The model's performance is limited by the quality and representativeness of the training data. - It may not generalize well to different types of toxic comments or diverse datasets. - Model performance can be impacted by changes in language trends over time.

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References:

[1] S. Saxena, "LSTM | Introduction to LSTM | Long Short Term Memor," Analytics Vidhya, Mar. 16, 2021. https://www.analyticsvidhya.com/blog/2021/03/introduction-to-long-short-term-memory-lstm/

[2] N. Donges, “Recurrent neural networks 101: Understanding the basics of RNNs and LSTM,” Built In, 2019. https://builtin.com/data-science/recurrent-neural-networks-and-lstm

[3] P. Srivastava, “Essentials of Deep Learning: Introduction to Long Short-Term Memory,” Analytics Vidhya, Dec. 10, 2017. https://www.analyticsvidhya.com/blog/2017/12/fundamentals-of-deep-learning-introduction-to-lstm/#h-4-architecture-of-lstms (accessed Dec. 22, 2023).

[4] “TensorFlow vs Keras - Javatpoint,” www.javatpoint.com. https://www.javatpoint.com/tensorflow-vs-keras (accessed Dec. 22, 2023).

[5] K. Hajian-Tilaki, “Receiver Operating Characteristic (ROC) Curve Analysis for Medical Diagnostic Test Evaluation,” Caspian Journal of internal medicine, vol. 4, no. 2, pp. 627–35, 2013, Available: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3755824/

[6] S. Narkhede, “Understanding AUC - ROC Curve,” Medium, Jun. 27, 2018. https://towardsdatascience.com/understanding-auc-roc-curve-68b2303cc9c5

[7] https://web.stanford.edu/class/archive/cs/cs224n/cs224n.1184/reports/6837517.pdf

[8] Pinkesh Badjatiya, Shashank Gupta, Manish Gupta, and Vasudeva Varma. Deep learning for hate speech detection in tweets. In Proceedings of the 26th International Conference on World Wide Web Companion, pages 759–760, 2017.

[9] Alexandre Ashade Lassance Cunha, Melissa Carvalho Costa, and Marco Aur´elio C Pacheco. Sentiment analysis of YouTube video comments using deep neural networks. In International Conference on Artificial Intelligence and Soft Computing, pages 561–570. Springer, 2019.

[10] M. Tocoglu and A. Onan, “A term weighted neural language model and stacked bidirectional LSTM based framework for sarcasm identification,” IEEE Access, vol. 9, pp. 7701–7722, 2021.

[10] A. Onan, “Sentiment analysis on massive open online course evaluations: a text mining and deep learning approach,” Computer Applications in Engineering Education, vol. 29, no. 3, pp. 572–589, 2020.

[11] K. Poojitha, A. Charish, M. Arun, K. Reddy, and S. Ayyasamy, “Classification of social media Toxic comments using Machine learning models.” Available: https://arxiv.org/pdf/2304.06934

[12] T. Mandl, S. Modha, P. Majumder, M. Dave, D. Patel, C. Mandlia, Jaiswal, A. Patel, Hasoc(2020) at fire 2019: Hate speech and offensive content identification in Indo-European languages., in: Overview of the HASOC track at FIRE 2019: Hate Speech and Offensive Content Identification in Indo-European Languages, 2019.

[13] R. Kumar, A. K. Ojha, S. Malmasi, M. Zampieri, Benchmarking aggression identification in social media, in: Proceedings of the First Workshop on Trolling, Aggression and Cyberbullying (TRAC-2018), 2018, pp. 1–11. https://aclanthology.org/W18-4401/

[14] L. Page, “The Sigmoid in Regression, Neural Network Activation and LSTM Gates,” Github.io, Jan. 17, 2020. https://suzyahyah.github.io/machine%20learning/2020/01/17/Sigmoid.html (accessed Dec. 27, 2023).

[15] What is the intuition of using tanh in LSTM?” Stack Overflow. https://stackoverflow.com/questions/40761185/what-is-the-intuition-of-using-tanh-in-lstm (accessed Dec. 27, 2023).

[16] J. Brownlee, “Difference Between Return Sequences and Return States for LSTMs in Keras,” Machine Learning Mastery, Oct. 23, 2017. https://machinelearningmastery.com/return-sequences-and-return-states-for-lstms-in-keras

[17] “9.2. Long Short-Term Memory (LSTM) — Dive into Deep Learning 0.14.4 documentation,” d2l.ai. https://d2l.ai/chapter_recurrent-modern/lstm.html

[18] “LSTM Architecture | Understanding the LSTM Architecture,” Analytics Vidhya, Jan. 21, 2021. https://www.analyticsvidhya.com/blog/2021/01/understanding-architecture-of-lstm/

[19] “Understanding of LSTM Networks,” GeeksforGeeks, May 10, 2020. https://www.geeksforgeeks.org/understanding-of-lstm-networks/

[20] S. Yan, "Understanding LSTM and its diagrams," Medium, Nov. 15, 2017. https://blog.mlreview.com/understanding-lstm-and-its-diagrams-37e2f46f171